

Tips and Tools for Protecting Networks from Security Breaches

Networks are prone to viruses and hacking. Internet users need to be aware of what threats and risks they are susceptible to, and what evasive and protective measures they can take to secure themselves. You will get an idea about the different network security techniques, as well as the vulnerabilities and the top open source tools to overcome them, from this article.



Computer security breaches have become common nowadays, with several occurring around the world every day. Some of the breaches are considered minor, with little loss of monetary or data resources, but many of them are major, or even catastrophic. Many of us have heard of the hacking of Sony Pictures' network in 2014, which the FBI attributed to the government of North Korea. More recently, hackers from Russia have been accused of tampering with the voting systems in the 2016 US presidential elections, and also with the data of Ukrainian businesses and several government agencies. So hackers are continuously looking out for new vulnerabilities to exploit. When our networks are not secure, then the information about individuals,

organisations, and even our government, is at the risk of being exposed or leveraged against us.

Many of us use artificial intelligence (AI) and the IoT in one way or the other to make our lives smarter – be it for home automation or industrial automation. Hackers have an eagle eye on the data generated by IoT devices. They try to enter and invade networks through this medium and ultimately steal whole sets of data or infect the complete network. Hence, network security is something that cannot be taken for granted.

According to the US based SANS Institute, network security is the process of adopting different preventive measures to protect the underlying network infrastructure